FN3 Worked Examples: Reading the Graph of a Function

Recording companion for the FN3 CheckIt support video — Math 172, Santa Ana College

How to use this document

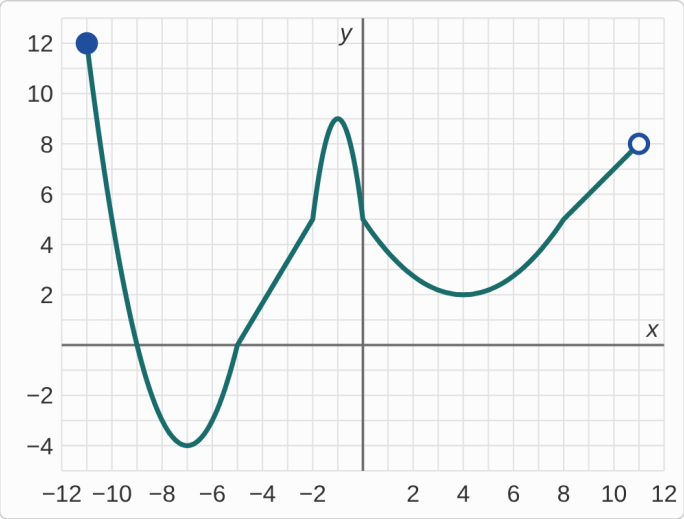
- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Is this question about x-values, read left to right, or y-values, read bottom to top? Domain and intervals of increase or decrease are x-values. Range, maxima and minima are y-values, and a maximum or minimum is located at a point. At each end of the graph, a filled dot means the end is included (a bracket) and an open dot means it is not (a parenthesis).

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Example 1: Domain



Plotted data: A graph of g on a grid with lines every 1 unit, with x from −12 to 12 and y from −5 to 13. From a filled dot at (−11, 12), it goes down steeply through (−9, 0) to (−7, −4), where it turns. It goes up through (−5, 0) to (−1, 9), where it turns, then down through (0, 5) to (4, 2), where it turns again. Then it goes up to an open dot at (11, 8). There are small corners where its pieces join, at x = −5, −2, 0 and 8.

Question

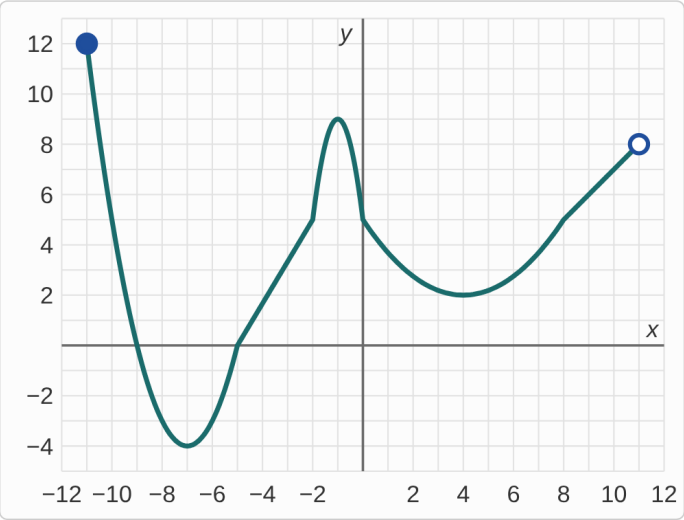
What is the domain of g?

- **A.** [−11, 11]
- **B.** [−11, 11)
- **C.** (−9, 11)
- **D.** [−12, 12]

Answer and reasoning

The domain is every *x*-value the graph covers, read left to right. It starts at $x=-11$ with a filled dot, so −11 is included: a bracket. It ends at $x=11$ with an open dot, so 11 is left out: a parenthesis. *[−11,11]* ignores the open dot; *(−9,11)* starts at the first *x*-intercept instead of the left end; and *[−12,12]* reads the edges of the grid instead of the graph. ***[−11,11)*.**

Example 2: Range



Plotted data: A graph of g on a grid with lines every 1 unit, with x from −12 to 12 and y from −5 to 13. From a filled dot at (−11, 12), it goes down steeply through (−9, 0) to (−7, −4), where it turns. It goes up through (−5, 0) to (−1, 9), where it turns, then down through (0, 5) to (4, 2), where it turns again. Then it goes up to an open dot at (11, 8). There are small corners where its pieces join, at x = −5, −2, 0 and 8.

Question

What is the range of g?

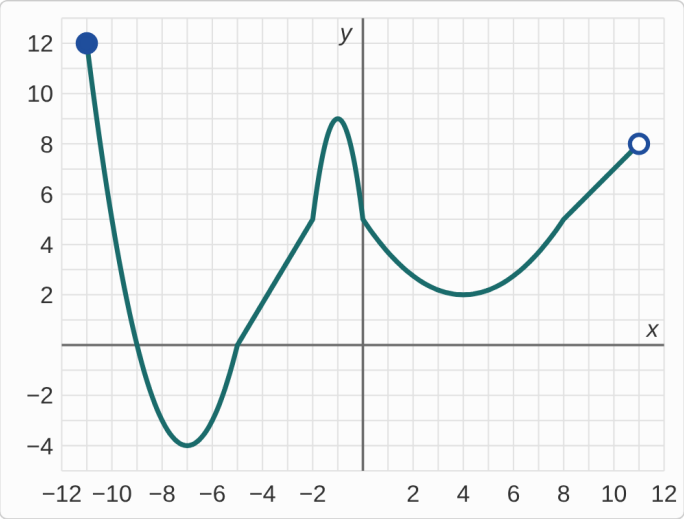
A. [−4, 9]
B. [−4, 8)
C. [−4, 12)
D. [−4, 12]

Answer and reasoning

The range is every *y*-value the graph reaches, bottom to top. The lowest is −4, at *(−7,−4)*. The highest is not the peak at 9: the left end, *(−11,12)*, is higher, and its dot is filled, so 12 is reached. *[−4,9]* stops at the highest turning point; *[−4,8)* uses the height of the right end, which is lower than the left end; and *[−4,12)* borrows the open dot from the other end, which is at height 8, not 12. ***[−4,12]*.**

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Example 3: x-Intercepts



Plotted data: A graph of g on a grid with lines every 1 unit, with x from −12 to 12 and y from −5 to 13. From a filled dot at (−11, 12), it goes down steeply through (−9, 0) to (−7, −4), where it turns. It goes up through (−5, 0) to (−1, 9), where it turns, then down through (0, 5) to (4, 2), where it turns again. Then it goes up to an open dot at (11, 8). There are small corners where its pieces join, at x = −5, −2, 0 and 8.

Question

What are the x-intercepts of g?

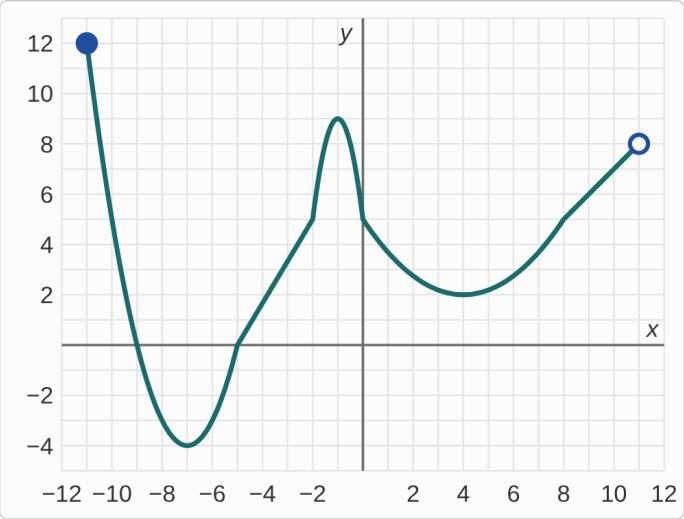
A. (−9, 0) and (−5, 0)
B. (0, −9) and (0, −5)
C. (−7, 0) and (4, 0)
D. (−11, 0) and (−5, 0)

Answer and reasoning

x-intercepts are the points where the graph meets the *x*-axis, where *y=0*. This graph meets it twice, crossing it each time, at *(−9,0)* and *(−5,0)*, and stays above the axis everywhere to the right of −5. *(0,−9)* and *(0,−5)* swap the coordinates; *(−7,0)* and *(4,0)* use the *x*-values of the low turning points, which are not on the axis; and the graph starts at *x=−11* at height 12, not on the axis. ***(−9,0)* and *(−5,0)*.**

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Example 4: y-Intercept



Plotted data: A graph of g on a grid with lines every 1 unit, with x from −12 to 12 and y from −5 to 13. From a filled dot at (−11, 12), it goes down steeply through (−9, 0) to (−7, −4), where it turns. It goes up through (−5, 0) to (−1, 9), where it turns, then down through (0, 5) to (4, 2), where it turns again. Then it goes up to an open dot at (11, 8). There are small corners where its pieces join, at x = −5, −2, 0 and 8.

Question

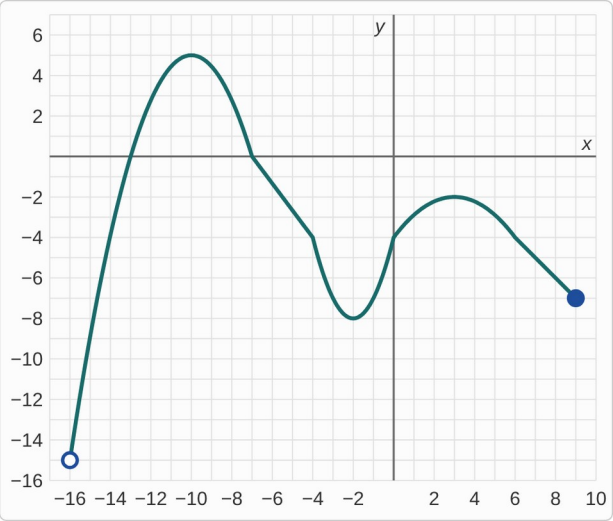
What is the y-intercept of g?

A. (0, 5)
B. (5, 0)
C. (0, 9)
D. (−1, 9)

Answer and reasoning

The *y*-intercept is the point where the graph meets the *y*-axis, where *x=0*. A function has at most one. There the graph is at height 5, so *(0,5)*. *(5,0)* swaps the coordinates; *(0,9)* and *(−1,9)* use the nearby peak, which is just left of the axis, not on it. ***(0,5).***

Example 5: Decreasing



Plotted data: A graph of g on a grid with lines every 1 unit, with x from −17 to 10 and y from −16 to 7. From an open dot at (−16, −15), it goes up steeply through (−13, 0) to (−10, 5), where it turns. It goes down through (−7, 0) to (−2, −8), where it turns, then up through (0, −4) to (3, −2), where it turns again. Then it goes down to a filled dot at (9, −7). There are small corners where its pieces join, at x = −7, −4, 0 and 6.

Question

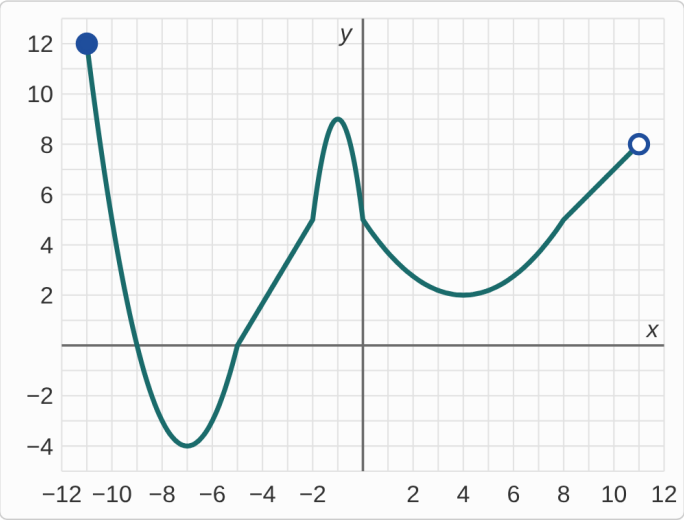
On which intervals is g decreasing?

A. (−8, 5) ∪ (−7, −2)
B. (−10, −2) ∪ (3, 9]
C. (−10, −2) ∪ (3, 9)
D. (−10, 3)

Answer and reasoning

Read left to right and keep the *x*-values where the graph goes down: from the turn at *x=−10* to the turn at *x=−2*, and from the turn at *x=3* to the right end, *x=9*. Intervals of increase and decrease are written with parentheses, even at a filled dot: CheckIt writes *(3,9)* although 9 is in the domain. *(−8,5)∪(−7,−2)* uses heights; *(3,9]* copies the bracket from the domain; and *(−10,3)* runs across a stretch where the graph goes up.
(−10,−2)∪(3,9).

Example 6: Local Minima



Plotted data: A graph of g on a grid with lines every 1 unit, with x from −12 to 12 and y from −5 to 13. From a filled dot at (−11, 12), it goes down steeply through (−9, 0) to (−7, −4), where it turns. It goes up through (−5, 0) to (−1, 9), where it turns, then down through (0, 5) to (4, 2), where it turns again. Then it goes up to an open dot at (11, 8). There are small corners where its pieces join, at x = −5, −2, 0 and 8.

Question

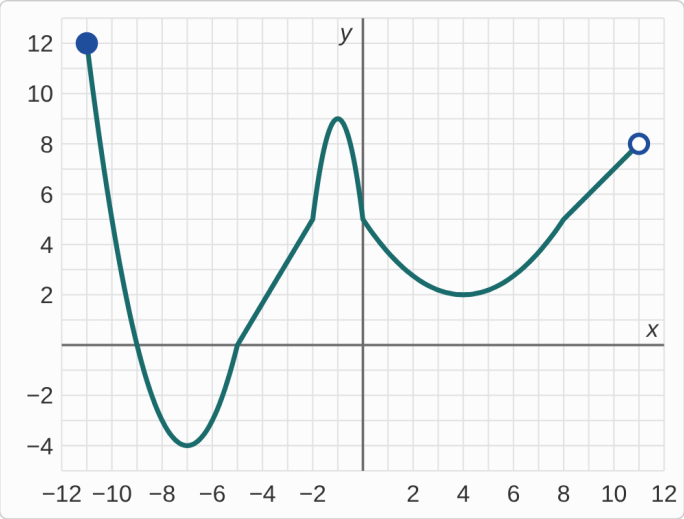
What are the local minima of g?

A. (−7, −4) only, the lowest point
B. (−4, −7) and (2, 4)
C. (−7, −4) and (4, 2)
D. (−11, 12) and (11, 8), the ends

Answer and reasoning

A local minimum is a low point compared with its neighbors on both sides: the graph comes down to it and goes back up. That happens at *(−7,−4)* and at *(4,2)*. *(4,2)* counts even though it is not the lowest point overall: that would be the global minimum. The ends are not local extrema, since the graph does not continue past them. *(−4,−7)* and *(2,4)* swap the coordinates. Each point gives both the minimum value and where it occurs: at *(−7,−4)*, the local minimum value is −4, occurring at $x=-7$. ***(−7,−4)* and *(4,2)*.**

Example 7: Global Maximum



Plotted data: A graph of g on a grid with lines every 1 unit, with x from −12 to 12 and y from −5 to 13. From a filled dot at (−11, 12), it goes down steeply through (−9, 0) to (−7, −4), where it turns. It goes up through (−5, 0) to (−1, 9), where it turns, then down through (0, 5) to (4, 2), where it turns again. Then it goes up to an open dot at (11, 8). There are small corners where its pieces join, at x = −5, −2, 0 and 8.

Question

What is the global maximum of g?

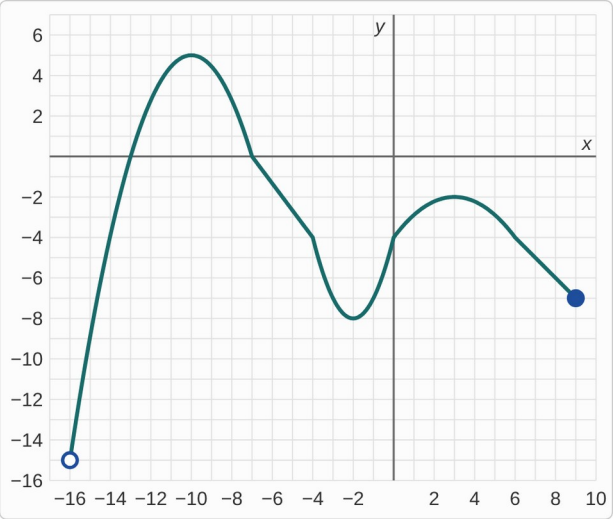
- **A.** 9, at the peak (−1, 9)
- **B.** 8, at the right end
- **C.** There is none, since 12 is at an end
- **D.** 12, at the left end (−11, 12)

Answer and reasoning

The global maximum is the greatest *y*-value the graph reaches anywhere. The left end, *(−11,12)*, has a filled dot, so the graph reaches 12, higher than the peak at 9. An end can be a global maximum when its dot is filled; it just cannot be a local one. 9 is the local maximum; 8 is at the right end, which is lower and also has an open dot. **12, at the left end.**

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Example 8: Global Minimum



Plotted data: A graph of g on a grid with lines every 1 unit, with x from −17 to 10 and y from −16 to 7. From an open dot at (−16, −15), it goes up steeply through (−13, 0) to (−10, 5), where it turns. It goes down through (−7, 0) to (−2, −8), where it turns, then up through (0, −4) to (3, −2), where it turns again. Then it goes down to a filled dot at (9, −7). There are small corners where its pieces join, at x = −7, −4, 0 and 6.

Question

What is the global minimum of g?

- **A.** −15, at the left end
- **B.** There is none.
- **C.** −8, at the low point (−2, −8)
- **D.** −7, at the right end

Answer and reasoning

The graph falls toward *(−16,−15)* on the left, but that dot is open, so it never reaches −15. It gets as close as you like: −14.9, −14.99, and so on, with no lowest value. So there is no global minimum, and the range is *(−15,5]*. −15 itself is not reached; −8 is a local minimum, but the graph goes lower near the left end; and −7 is not the lowest value. **There is none.**

Example 9: From a List

$$(-9,-4)\cup(1,6)$$

Question

Your graph must be decreasing on (−9, −4) ∪ (1, 6) and increasing between those intervals. What must it have at x = −4?

A. A local minimum
B. A local maximum
C. An x-intercept
D. A vertical asymptote

Answer and reasoning

Going left to right, the graph is decreasing until $x=-4$ and then increasing, so $x=-4$ is the bottom of a valley: a local minimum. By the same reasoning it has a local maximum at $x=1$, where increasing turns to decreasing. A local maximum would need increasing then decreasing; the list says nothing about where the graph crosses the x-axis; and these graphs are unbroken curves. **A local minimum.**

Example 10: The Ends

$$(-8,10]$$

Question

Your graph must have domain (−8, 10]. How should its ends look?

- **A.** Filled dots at both ends, x = −8 and x = 10
- **B.** Open dots at both ends, x = −8 and x = 10
- **C.** Open dot at x = −8, filled dot at x = 10
- **D.** Arrows at both ends, since it is an interval

Answer and reasoning

A parenthesis means the end is left out: an open dot at $x=-8$. A bracket means it is included: a filled dot at $x=10$. The graph goes all the way across between them. Filled dots at both ends would be $[-8,10]$, and open dots at both ends would be $(-8,10)$. Arrows would mean the graph keeps going forever, but this domain stops at both ends.
An open dot at x = −8 and a filled dot at x = 10.

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Example 11: The Other Direction

Table 1. The feature list for a sketch

Domain	(−10, 9]
Local maxima	(−8, 4) and (3, −1)
Local minimum	(−2, −6)
Decreasing on	(−8, −2) ∪ (3, 9)

Question

Like a CheckIt list, this one does not say where the graph increases. Where must it increase?

A. (−2, 3) only, between the turns
B. (−10, −8) ∪ (−2, 3)
C. (−8, −2) ∪ (3, 9)
D. (−6, 4)

Answer and reasoning

Everywhere in the domain that the graph is not decreasing, it is increasing, apart from the turning points themselves. The domain starts at −10 and the graph is decreasing from −8, so it rises from the left end up to the local maximum at $x=-8$. It also rises from the local minimum at $x=-2$ to the local maximum at $x=3$. $(-2,3)$ alone forgets the stretch from the left end; $(-8,-2)\cup(3,9)$ repeats the decreasing intervals; and $(-6,4)$ uses heights. ***$(-10,-8)\cup(-2,3)$*.**

Example 12: Impossible?

Table 2. Four lists of features for a sketch

List A	Local max at (−3, 6), local min at (2, 1), decreasing on (−3, 2)
List B	y-intercept (0, 4), local min at (2, 1), decreasing on (0, 2)
List C	Local min at (1, 5), local max at (4, 2), increasing on (1, 4)
List D	x-intercepts (−5, 0) and (−1, 0), local min at (−3, −2)

Question

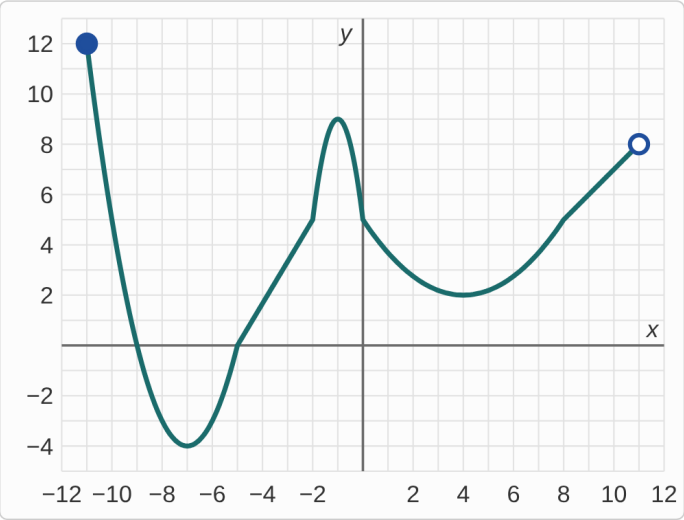
Which list cannot all be true for the same graph?

A. List A
B. List B
C. List C
D. List D

Answer and reasoning

In List C, increasing on *(1,4)* means the graph is higher at $x=4$ than at $x=1$. But the list puts it at height 5 at $x=1$ and height 2 at $x=4$, lower. A graph cannot rise and end up lower. The others can all happen: List A goes down from 6 to 1; List B crosses the y-axis at 4 and goes down to 1; and List D has a valley at −2 between two crossings of the x-axis. **List C.**

Example 13: Slip 1



Plotted data: A graph of g on a grid with lines every 1 unit, with x from −12 to 12 and y from −5 to 13. From a filled dot at (−11, 12), it goes down steeply through (−9, 0) to (−7, −4), where it turns. It goes up through (−5, 0) to (−1, 9), where it turns, then down through (0, 5) to (4, 2), where it turns again. Then it goes up to an open dot at (11, 8). There are small corners where its pieces join, at x = −5, −2, 0 and 8.

Question

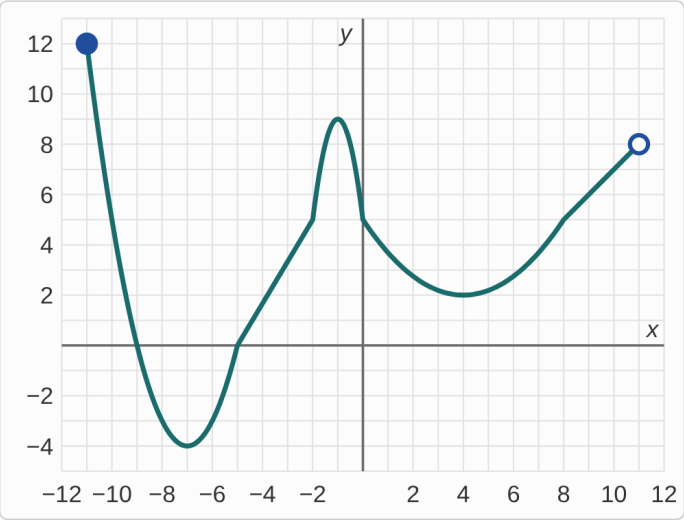
A student says g is increasing on $(-4, 9) \cup (2, 8)$. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** Those are the intervals where g decreases, not where it increases.
- **C.** Those are y-values; use x-values: $(-7, -1) \cup (4, 11)$.
- **D.** Intervals of increase are written with brackets, not parentheses.

Answer and reasoning

The student read heights: −4 up to 9, and 2 up to 8. Intervals of increase are written with *x*-values: the graph rises from the turn at $x=-7$ to the turn at $x=-1$, and from the turn at $x=4$ to the right end at $x=11$. These are the rising stretches, not the falling ones, and CheckIt writes them with parentheses. **Increasing on $(-7,-1)\cup(4,11)$.**

Example 14: Slip 2



Plotted data: A graph of g on a grid with lines every 1 unit, with x from −12 to 12 and y from −5 to 13. From a filled dot at (−11, 12), it goes down steeply through (−9, 0) to (−7, −4), where it turns. It goes up through (−5, 0) to (−1, 9), where it turns, then down through (0, 5) to (4, 2), where it turns again. Then it goes up to an open dot at (11, 8). There are small corners where its pieces join, at x = −5, −2, 0 and 8.

Question

A student says g has local maxima at (−1, 9) and (−11, 12). What is the mistake, if any?

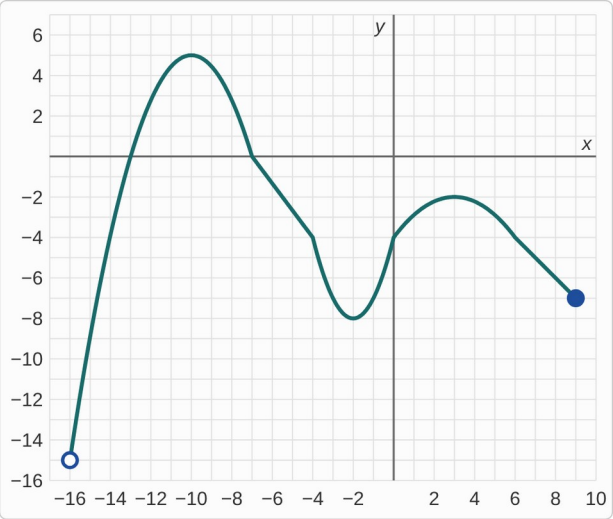
A. There is no mistake.
B. The end (−11, 12) is not a local max; only (−1, 9) is.
C. Only (−11, 12) counts, since it is the highest point on the graph.
D. Local maxima are y-values, so the answer should be 9 and 12.

Answer and reasoning

A local maximum is a peak: the graph rises to it and then falls, so there is graph on both sides. At the left end there is nothing to its left, so CheckIt does not count it as a local maximum, even though it is the global maximum. The peak at *(−1,9)* counts even though it is not the highest point overall: its local maximum value is 9, occurring at *x=−1*. Answering “9 and 12” keeps the endpoint and drops where each value occurs;

CheckIt names a local maximum by its point, as in “local max at $(-1,9)$.” **The only local maximum is at $(-1,9)$.**

Example 15: Slip 3



Plotted data: A graph of g on a grid with lines every 1 unit, with x from −17 to 10 and y from −16 to 7. From an open dot at (−16, −15), it goes up steeply through (−13, 0) to (−10, 5), where it turns. It goes down through (−7, 0) to (−2, −8), where it turns, then up through (0, −4) to (3, −2), where it turns again. Then it goes down to a filled dot at (9, −7). There are small corners where its pieces join, at x = −7, −4, 0 and 6.

Question

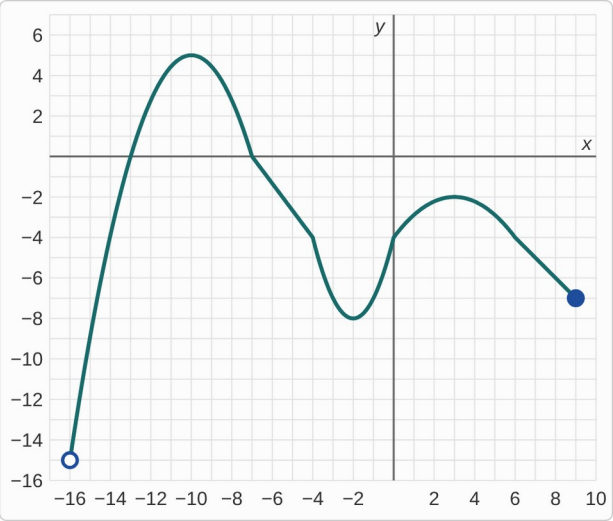
A student says the domain of g is (−16, 9]. What is the mistake, if any?

A. There is no mistake.
B. It should be [−16, 9], since the graph starts at −16.
C. It should be (−17, 10), the edges of the grid shown.
D. It should be (−13, −7), between the x-intercepts.

Answer and reasoning

The graph runs from $x=-16$, with an open dot, so a parenthesis, to $x=9$, with a filled dot, so a bracket. *(−16,9]* is right. A bracket at −16 would ignore the open dot; *(−17,10)* reads the edges of the grid instead of the graph; and the domain covers the whole graph, not just the stretch between the x-intercepts. **There is no mistake.**

Example 16: Slip 4



Plotted data: A graph of g on a grid with lines every 1 unit, with x from −17 to 10 and y from −16 to 7. From an open dot at (−16, −15), it goes up steeply through (−13, 0) to (−10, 5), where it turns. It goes down through (−7, 0) to (−2, −8), where it turns, then up through (0, −4) to (3, −2), where it turns again. Then it goes down to a filled dot at (9, −7). There are small corners where its pieces join, at x = −7, −4, 0 and 6.

Question

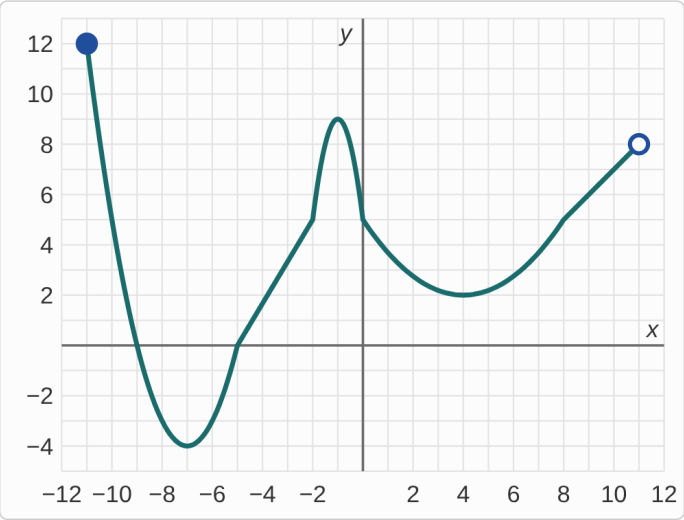
A student says the range of g is [−8, 5]. What is the mistake, if any?

A. There is no mistake.
B. It should be [−16, 9], running from the left end to the right end.
C. It should be [−15, 5], since the graph starts at a height of −15.
D. It dips below −8 at the left: the range is (−15, 5].

Answer and reasoning

The lowest turning point is −8, but the graph keeps going down toward the open dot at *(−16,−15)*, so it takes every height above −15 without reaching −15. At the top, the highest turning point is reached, so that end gets a bracket. *[−16,9]* uses the *x*-values of the ends, not heights, and a bracket at −15 ignores the open dot. ***(−15,5]*.**

Example 17: Slip 5



Plotted data: A graph of g on a grid with lines every 1 unit, with x from −12 to 12 and y from −5 to 13. From a filled dot at (−11, 12), it goes down steeply through (−9, 0) to (−7, −4), where it turns. It goes up through (−5, 0) to (−1, 9), where it turns, then down through (0, 5) to (4, 2), where it turns again. Then it goes up to an open dot at (11, 8). There are small corners where its pieces join, at x = −5, −2, 0 and 8.

Question

A student says g has no global minimum, because the graph ends at an open dot. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** The open dot is not the lowest place; the global min is −4.
- **C.** The global minimum is 8, at the open dot on the right end.
- **D.** The global minimum is 2, at the last turning point, (4, 2).

Answer and reasoning

An open dot only rules out a global maximum or minimum when it is at the very top or bottom of the graph. Here the open dot at *(11,8)* is partway up, and the graph is lower elsewhere: its lowest point is *(−7,−4)*, which is reached. So the global minimum is −4. The open dot's height, 8, is not reached and is not the lowest; and 2 is a local minimum, but the graph goes lower at *x*=−7. **The global minimum is −4.**